Total No. of Questions : 5] [Total No. of Printed Pages : 5

Roll No. ____________________

REF NO. 023513

B. Sc. (Hons) Semester V Examination
2 0 1 9 – 2 0

PHYSICS

Paper No. BPT-505 : Electromagnetic Theory

Time : 3 hours **Full Marks : 70**

(Write your Roll No. at the top immediately on the receipt of this question paper)

The figures in the right-hand margin indicate marks.
Answer all questions
All symbols have their usual meanings unless specified otherwise

1. **Answer any four of the following :** **4×5**

   (a) Obtain the electric field inside and outside of a long hollow cylinder which carries a uniform surface charge (σ).

   (b) If vector function $\vec{F} = \hat{a}_x(3y - c_1 z) + \hat{a}_y(c_2 x - 2z) - \hat{a}_z(c_3 y + z)$ is irrotational, determine the constants c_1, c_2 and c_3.

   (c) Why does any net charge resides on the surface of a conductor? Explain.

   (d) If $\vec{P} = 2\sin(10t + x - \pi/4)\hat{a}_y$ and $\vec{Q}_s = e^{jk}(\hat{a}_x - \hat{a}_z)\sin(\pi y)$, determine the phasor form of $\vec{P}$ and the instantaneous form of $\vec{Q}_s$.

   (e) A plane e.m. wave with perpendicular polarization is incident from air on water surface (dielectric constant $\varepsilon_r = 80$). Compute the Brewster angle ($\theta_{B\perp}$) and corresponding angle of transmission.

   (f) Compute the loss tangent and dielectric constant of a non-magnetic medium whose intrinsic impedance is given by $240\angle 30^\circ\ \Omega$.

— **(2)** —

2. (a) Consider a parallel plate capacitor immersed in seawater carries a voltage $V_0 e^{j\omega t}$. If the seawater is characterized by a permittivity $\varepsilon = 81\varepsilon_0$ and permeability $\mu = \mu_0$, resistivity $\rho = 0.23\ \Omega \cdot \text{m}$ at frequency $\omega = 2\pi \times 10^8$ Hz. What is the ratio of conduction current to displacement current? **5**

   (b) If $V = 0$ and $\vec{A} = A_0 \sin(kx - \omega t)\hat{a}_y$ (A_0, ω and k are parameters), calculate $\vec{E}$ and $\vec{B}$. Do $\vec{E}$ and $\vec{B}$ satisfy Maxwell's equation in vacuum? If so, how are ω and k are related? **7**

— **(3)** —

3. (a) The instantaneous electric field of an e.m. wave is given by

$$\vec{E} = \hat{a}_x E_{x0} \cos(\omega t - \beta z + \phi_x) + \hat{a}_y E_{y0} \cos(\omega t - \beta z + \phi_y)$$

If $\phi_x = +\pi/2, \phi_y = 0$ and $E_{x0} = E_{y0} = E_R$, determine the polarization and sense of rotation of the wave. **5**

(b) Discuss the e.m. wave propagation in good conductor and show that the magnetic field intensity lags behind the electric field intensity by 45°. **7**

— (4) —

4. (a) The general solution of the Laplace equation in spherical polar coordinate with azimuthal symmetry is on the form

$$V(r, \theta) = \sum_{l=0}^{\infty} \left(A_l r^l + \frac{B_l}{r^{l+1}} \right) P_l(\cos\theta)$$

Determine the potential outside of a sphere of radius R subject to boundary condition $V(R, \theta) = \cos\theta$. **7**

(b) In electrostatics, show that the boundary conditions for the electrostatic fields show discontinuity whereas that of potentials are continuous. **5**

Or

(a) Does the Poynting vector correspond to dual character in the energy momentum conservation in electrodynamics? If so, explain. **6**

(b) What is linear medium? How does the Maxwell's equation modify in such a linear dielectric? **6**

5. (a) Prove that for transverse electric mode in rectangular waveguide, the expression for z-component of magnetic field is given by

$$H_{zs} = H_0 \cos\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) e^{-\gamma z}$$

6

(b) A rectangular waveguide with dimensions $a = 2.5$ cm, $b = 1.0$ cm is to operate below 15.1 GHz. How many TE and TM modes can the waveguide transmit if the guide is filled with a medium characterized by $\sigma = 0, \varepsilon = 4\varepsilon_0, \mu_r = 1$? Calculate the cutoff frequencies of the modes. **8**

Or

(a) Show that in the case of perpendicular polarization, the Brewster angle is given by

$$\sin^2(\theta_{B\perp}) = \frac{1 - \mu_1\varepsilon_2/\mu_2\varepsilon_1}{1 - (\mu_1/\mu_2)^2}$$

7

(b) When a uniform plane e.m. wave is incident on a plane conducting surface (medium 2, $\sigma = \infty$) from a lossless dielectric (medium 1, with $\sigma = 0$) obliquely, find the expression for total electric and magnetic fields in medium 1 for perpendicular polarization case. **7**

Parameters to be used in the calculations :

Given $\mu_0 = 4\pi \times 10^{-7}$ H/m, $\varepsilon_0 = 8.854 \times 10^{-12}$ F/m

Charge of electron, $e = 1.6 \times 10^{-19}$ Coulomb

***
